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Volcanic lake systematics II. Chemical constraints

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Abstract

A database of 373 lake water analyses from the published literature was compiled and used to explore the geochemical systematics of volcanic lakes. Binary correlations and principal component analysis indicate strong internal coherence among most chemical parameters. Compositional variations are influenced by the flux of magmatic volatiles and/or deep hydrothermal fluids. The chemistry of the fluid entering a lake may be dominated by a high-temperature volcanic gas component or by a lower-temperature fluid that has interacted extensively with volcanic rocks. Precipitation of minerals like gypsum and silica can strongly affect the concentrations of Ca and Si in some lakes. A much less concentrated geothermal input fluid provides the mineralized components of some more dilute lakes. Temporal variations in dilution and evaporation rates ultimately control absolute concentrations of dissolved constituents, but not conservative element ratios.

Most volcanic lake waters, and presumably their deep hydrothermal fluid inputs, classify as immature acid fluids that have not equilibrated with common secondary silicates such as clays or zeolites. Many such fluids may have equilibrated with secondary minerals earlier in their history but were re-acidified by mixing with fresh volcanic fluids. We use the concept of 'degree of neutralization' as a new parameter to characterize these acid fluids. This leads to a classification of gas-dominated versus rock-dominated lake waters. A further classification is based on a cluster analysis and a hydrothermal speedometer concept which uses the degree of silica equilibration of a fluid during cooling and dilution to evaluate the rate of fluid equilibration in volcano-hydrothermal systems.

Keywords: volcanic lake water compositions; cluster analysis; degree of neutralization

1. Introduction

Volcanic lakes range in composition from very dilute, meteoric waters to hyper-acid brines. Variations in lake-water chemistry are related to variations in the composition and flux of volcanic fluids and gases into the lake, with superimposed dilution and evaporation effects. Long-term monitoring at Crater Lake on Mt. Ruapehu (e.g. Giggenbach, 1974; Christenson and Wood, 1993; Christenson, 2000 – this volume), Laguna Caliente at Poas Volcano (e.g. Brantley et al., 1987; Rowe et al., 1992a,b; Martinez et al., 2000 – this volume), Yugama Lake at Kusatsu-Shirane Volcano (e.g. Takano and Watanuki, 1990 and Ohba et al., 1994, 2000 – this volume), and Lake Nyos in Cameroon (e.g. Kusakabe et al., 1989, 2000 – this volume; Evans et al., 1993) have documented the wide range of physical and chemical changes that can occur in volcanic and Crater Lakes. Such changes are believed to reflect water–rock (WR) interactions that occur in and beneath the lake and variations in the flux of heat and volatiles derived from subsurface hydrothermal/magmatic systems. Many volcanic lakes represent dynamic hydrologic systems with high recharge rates and a rapid circulation of fluids through the volcanic edifice resulting

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from the high fracture permeability of young volcanic rock sequences (Sanford et al., 1995). Volcanic lakes can thus be regarded as the surface expressions of volcano-hosted hydrothermal systems, and compositional characteristics of lake waters will respond to chemical and physical changes in the subsurface hydrothermal/magmatic system. For that reason, volcanic lakes are considered excellent environments for monitoring and in some cases, forecasting future volcanic activity (Giggenbach, 1974; 1983; Giggenbach and Glover, 1975; Rowe et al., 1992b; Badrudin, 1994; Takano et al., 2000a – this volume).

Giggenbach (1974) provided a genetic basis for classification of hydrothermal fluids, based on the degree of equilibration between fluids and secondary minerals. An extension of this approach (Giggenbach, 1990; Christenson and Wood, 1993) led to the recognition that the degree of WR interaction of a hyperacid volcanic brine has a spatial dimension that relates to the extent of the fluid layer between degassing magma and a lake. In some systems, cooling magma is almost intruding into the lake bottom, providing fresh volcanic rock to react with the escaping gases and lake waters (Giggenbach, 1974; Sigurdsson, 1977). In this regime, lake fluids are a mixture of meteoric water and quenched volcanic gas that are little modified by WR reaction. Other systems may have a volcanic gas input at several hundreds to thousands m below the lake bottom, and there an extensive sublimnic hydrothermal system exists (Rowe et al., 1992a; Christenson and Wood, 1993; Pasternack and Varekamp, 1994). The WR interaction in such a system may modify the volcanic gas phase substantially before entry into the lake or alternatively, sub-lake volcanics may have been altered to a mostly inert matrix by continuous circulation of acid fluids. Still other systems may experience diffuse emanation of highly diluted volcanic gases directly into lake bottom waters.

Pasternack and Varekamp (1997) proposed a classification scheme for volcanic lakes largely based on thresholds in the physical processes of the heat dissipation capacity of a lake and hydrodynamic mixing regimes. Qualitative considerations of volcanic lake processes and long-term data sets obtained at crater lakes on Mt. Ruapehu, Poas, and Kusatsu-Shirane volcanoes leave little doubt that geochemical and physical processes are strongly coupled. Those lakes subjected to direct degassing of high-temperature magmatic volatiles by near-surface magma bodies not only condense acidic brines, but also experience excess evaporation. In some cases, intense evaporation, in combination with climatic variations (e.g. rainfall), may upset the delicate hydrologic balance of such lakes, resulting in the complete disappearance of a lake (e.g. Poas, Rowe et al., 1992a).

We compiled a database of volcanic lake chemical compositions from the published literature and use it to explore the geochemical systematics of volcanic lakes. Multivariate statistical analysis is applied to the chemical data to obtain a classification of volcanic lakes. Other data are used to help identify the processes responsible for these groupings.

2. Data and methods

Chemical analyses of 373 volcanic lake fluids were compiled (Table 1). The data density is very irregular; long-term data sets are available from a few lakes such as Crater Lake, Mt. Ruapehu, Yugama, Mt. Kusatsu-Shirane, and Laguna Caliente, Poas, whereas limited temporal data are available from about ten other lakes. Monitoring of different lake types has involved different geochemical parameters, and in some studies physico-chemical data are reported in graphical formats only. As a result, data coverage varies from lake to lake, with HCO_3, Li, Mn, F, and B being the least commonly reported constituents (the lack of HCO_3 is related to the highly acidic nature of many lakes in the data base). Excluding analyses that lack these parameters leaves 140 complete entries. There is also little information regarding data precision and accuracy because samples were collected, preserved and analyzed using different methods. Furthermore, analyses of highly concentrated acid brines are subject to significant uncertainties, particularly for minor and trace constituents and even analyses of the same fluids in different labs can show large variations (Takano et al., 2000b – this volume). Direct contamination of hydrothermal fluids with seawater strongly influences compositions (e.g. Na concentrations) and lake water analyses with an obvious seawater component (e.g. Anak Krakatau) have been removed from the database.

Binary correlation coefficients of temperature and chemical data were calculated to characterize the

Table 1
Data base of lake water compositions[a]

Location	# of entries	Lakes; sampling periods
Central and South America (many Poas entries)	37	Poas '85–'91
Lake Yugama	32	Yugama '77–'89
Lake Ruapehu	86	Ruapehu '70–'96
Lake Nyos	48	Nyos '86–'92
Indonesian crater lakes (many Keli Mutu entries)	85	BA, EG, GA, KI, KM, KP, SR, WS,
Others	85	AL, AV, BN, BO, BR, CP, EC, EK, FO, FU, HE, HI, KA, MA, MN, MO, MS, NE, NB, NRA, NRT, OY, PP, RV, SC, TA, VI, WU

[a] AL, Lake Albano; AV, Lake Averno; BA, Batur Lake; BN, Bannoe; BO, Lake Bolsena; BR, Lake Bracciano; CP, Copahue; EC, El Chichon; EG, Egon Lake; EK, Ebeku-Kuriles; FO, Lake Fogo; FU, Lake Furnas; GA, Galunggung; HE, Herman Pit; HI, Hippopotamus pools; KA, Karymsky Lake; KI, Kawah Ijen; KM, Keli Mutu; KP, Kawah Putih; MA, Lake Martignano; MN, Lake Monoun; MO, Monticchio lakes; MS, Maly Semiachik; NE, Lake Nemi; NB, Noboribetsu; NRA, Nevado del Ruiz Agua Caliente; NRT, Nevado del Ruiz Termales del Ruiz; OY, Oyunuma; PP, Popocatepetl; RV, Rincon de la Vieja; SC, Lake Sete Cidades; SR, Sirung; TA, Taal; VI, Lake Vico; WS, Wai Sano; WU, Lake Wum.

A complete listing of sources and data can be obtained from the authors and also can be found on the volcanic lake web sites: Pasternack:http://lawr.ucdavis.edu/faculty/gpast/research.html; Varekamp:http://www.wesleyan.edu/ees/volclakes.html

relationships among various parameters for which data are available. Because compositional data for some common lake-water constituents span several orders of magnitude, we also calculated correlation coefficients for log-transformed data. Coefficients are divided into anion–anion, cation–cation, anion–cation, and global parameter–ion correlations to ascertain the overall relationships between volcanic gas inputs and rock-derived contributions, as well as relations between physical and chemical processes.

Principal components analysis (PCA) is a variable-oriented statistical technique that determines the minimum number of variables needed to explain the maximum amount of variance in a set of data; it does not require that data be normally distributed. Analyses were performed on both raw data and log-transformed data with the software package Statistica® v.98[1]. Of the determined principal components, only those with eigenvalues greater than one were retained, which means that the resulting components extract at least as much variance as one original variable. The results of PCA include percent of variation explained, communalities, loadings, and rotated loadings. Communality is an important statistic because it shows the proportion of variance of a constituent caused by retained principal components, so a communality greater than 0.9 indicates that virtually all of a variables' dynamics are explainable by PCA. A loading is the correlation between a lake constituent and a principal component, and a high loading identifies a principal component that is largely responsible for the variations of a lake constituent. Rotated loadings (we used the varimax normalized approach) maximize the differences between principal components so that it is easier to interpret the physical process underlying a component.

To differentiate volcanic lakes on the basis of their actual fluid geochemistry, the database was analyzed by use of cluster analysis, a sample-oriented, non-parametric statistical technique that groups data based on shared characteristics. The clustering algorithm we used was a hierarchical 'single linkage' scheme (Hair et al., 1992) within Statistica® v.98. The variables used to classify the water samples were the observed chemical constituents.

We first summarize some features of WR interaction in acid systems and then present the results of the statistical analyses and interpretation of these results. Results of the analyses are used to develop a broad classification scheme for volcanic lakes.

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3. Water–rock interaction processes in acid hydrothermal systems

Rock dissolution and WR reaction in volcanic lakes is probably limited because of the relatively low water temperatures and limited availability of rock surface (cf. Rowe et al., 1992a). Reaction of warm acid fluids with fine debris washed in during rainstorms may provide enough surface area for limited WR reactions to occur inside the lake to impact the overall lake composition. The WR ratios are much lower in the subsurface hydrothermal system than inside the lake, the temperatures tend to be higher, and intensive acid fluid-rock reaction takes place. Circulation of lake brines through the volcanic edifice is another mechanism for WR reaction of the acid lake waters (Rowe et al., 1992a).

Most early studies of hydrothermal fluids in neutral pH geothermal systems emphasized parameters that indicate the degree of reaction progress or equilibration with secondary minerals (Giggenbach, 1988). In such classifications, all our volcanic lake fluids fall within the 'immature group' in a Na/1000–K/100–$\sqrt{Mg}$ diagram and are far from equilibrium with secondary silicate phases. The Giggenbach (1988) approach of classification of geothermal fluids is not very suitable for application to volcano-hydrothermal systems, because that method is based on the equilibration of a starting fluid composition with a rock protolith to multiple saturation with secondary minerals. Most active volcano-hydrothermal systems have a continuous influx of volcanic gases that can re-acidify the partly neutralized fluids, and very few minerals will be saturated in these mixed fluids. So it becomes a question if the rate of acidification exceeds the rate of neutralization by WR reaction and the Giggenbach ternary diagram is not applicable when some of the salient parameters (e.g. pH) are externally determined (Chiodini et al., 1993).

Hydrothermal fluids with element ratios that are different from those in host rocks either have precipitated secondary phases during their history or must have formed by incongruent dissolution of volcanic rock during WR interaction. The main rock-derived elements in volcanic lake fluids from andesitic volcanoes are Na, Ca, K, Mg, Fe and Al, which are derived from the dissolution of volcanic glass, plagioclase, pyroxene and biotite or muscovite. We calculated the amounts of the 'easily leached' elements Na and Mg (following Rowe and Brantley, 1993) on a molar basis relative to the sum of major rock-forming elements (ΣRFE = Na + Ca + Mg + K + Al + Fe). Acid fluids that travel rapidly through a fresh volcanic rock matrix may have high (Na + Mg)/ΣRFE, whereas slow flowing fluids that have undergone more extensive reaction with fresh rocks would have lower (Na + Mg)/ΣRFE ratios. This parameter is also a function of the bulk composition of the rock matrix of the hydrothermal system, which tends to be andesitic in most systems. In addition, the reactivity of a fluid is influenced by TDS (total dissolved solids) levels as well. Hydrothermal systems that have already lost most of their easily-leachable cations during earlier phases of alteration should have fluids with low (Na + Mg)/$\sum$RFE as well.

Instead of using 'maturity of fluids' related to equilibration with secondary minerals, we consider here the 'degree of neutralization', a parameter that indicates the amount of acid consumption through rock dissolution but that does not specify the level of equilibration with individual rocks or minerals. The amount of H^+ consumed during WR interaction reflects the time-integrated WR history of the fluids and can be derived semi-quantitatively from the anion-H^+ balance. From the anion concentrations of volcanic lake fluids, we calculated the equivalent sum of anions (in μeq/g), which for pure strong acids should be equal to the H^+ concentration (μmol/g). We took into account that HSO_{4^-} is the dominant species at pH < 2 at moderate temperatures, whereas SO_4^{2-} dominates at pH > 2. We calculated the 'pure acid pH' that the fluid would have had if no WR interaction had occurred from the equivalent anion concentration. We then calculated the % of H^+ used (degree of neutralization) and % of H^+ left or percentage residual acidity (PRA) of the original H^+ reservoir, based on the measured and calculated H^+ quantities. An obvious shortcoming is the use of calculated H^+ *concentrations* from the anion analyses versus subtracted H^+ *activities* from the pH measurement. This has only a significant impact at pH values below pH ~ 0.5; correcting for it involves the proper activity–concentration relations for H^+ in hyper-acid brines, which is beyond the scope of this paper (Nordstrom and Alpers, 1998).

The saturation state of lake water with respect to primary and secondary minerals can be determined by use of an appropriate aqueous speciation model. We evaluated the aqueous speciation and saturation

Table 2
Correlation coefficients among volcanic lake chemical constituents. ∗ indicates $r > 0.7$ or $r < -0.7$

	T	pH	Cl	$S0_4$	F	B	Na	Mg	Al	SiO_2	K	Ca	Mn	Fe	Li
TDS	∗0.77	−0.50	∗0.90	∗0.98	∗0.83	0.17	0.41	0.68	∗0.86	0.32	∗0.82	0.55	0.54	∗0.83	−0.01
T	1.00	−0.48	∗0.71	∗0.74	0.66	−0.04	0.52	0.66	0.65	0.53	∗0.70	0.48	0.39	0.64	−0.22
pH		1.00	−0.48	−0.47	−0.34	−0.15	−0.24	−0.44	−0.46	−0.24	−0.39	−0.60	−0.57	−0.36	−0.14
Cl			1.00	∗0.80	∗0.85	0.24	0.41	0.62	∗0.76	0.38	∗0.71	0.59	∗0.70	∗0.85	0.00
SO_4				1.00	∗0.72	0.11	0.38	0.62	∗0.80	0.18	∗0.79	0.42	0.36	∗0.72	−0.01
F					1.00	0.46	0.67	0.55	∗0.83	0.36	∗0.83	0.42	0.28	∗0.99	0.25
B						1.00	0.23	0.03	0.17	−0.13	0.15	0.16	0.17	0.16	0.38
Na							1.00	0.61	0.31	0.08	0.61	0.27	0.46	0.34	0.06
Mg								1.00	0.63	0.47	0.64	0.63	∗0.92	0.59	−0.02
Al									1.00	0.21	∗0.83	0.43	0.33	∗0.82	−0.06
SiO_2										1.00	0.11	0.42	0.37	0.36	0.11
K											1.00	0.42	0.40	∗0.76	0.00
Ca												1.00	∗0.82	0.47	−0.01
Mn													1.00	0.36	0.29
Fe														1.00	0.01
Li															1.00

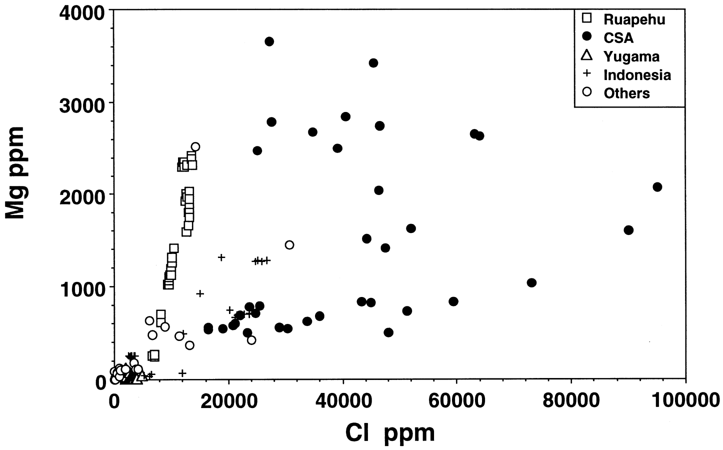

Fig. 1. Mg versus Cl for the various lake systems. At Ruapehu, small increases in Cl are accompanied by large Mg gains (intense WR interaction), whereas for a large part of the Poas data a strong increase in Cl with much lesser Mg gains is observed (dominance of volcanic gas component). CSA: samples from central and south American lakes.

state of many water samples from Table 1 with the computer program SOLVEQ (Reed, 1982) at the measured lake temperature (T_{lake}). Most of the WR interaction takes place in a higher temperature hydrothermal system below the lake, so computer simulations at T_{lake} are only a crude approximation. The choice of redox conditions has a strong influence on the saturation state of minerals that contain elements with multiple redox states (e.g. iron, manganese, carbon, sulfur). In some cases, it can be assumed that redox conditions in the volcanic lake are constrained by saturation with respect to native sulfur, but in many other cases, the exact Eh value is unknown and redox disequilibrium is to be expected. The saturation state of the many complex sulfide minerals is therefore beyond the scope of this overview, and the reader is referred to Rowe (1991), Christenson and Wood (1993), Delmelle and Bernard (1994), Pasternack and Varekamp (1994) and Sriwana et al. (2000 – this volume) for details. Also, lake waters may be more dilute or more concentrated (by surface evaporation) versions of deeper fluids, so that

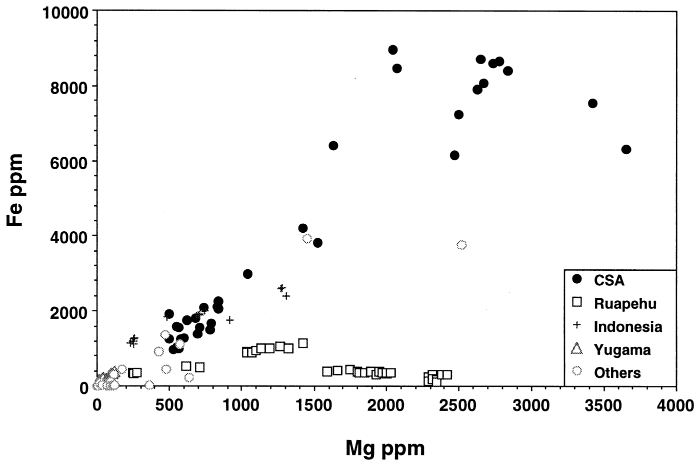

Fig. 2. Mg versus Fe for the lake waters: the Ruapehu data show a very flat array, suggesting precipitation of an Fe-bearing phase, whereas the Poas data show simultaneous increases in both elements.

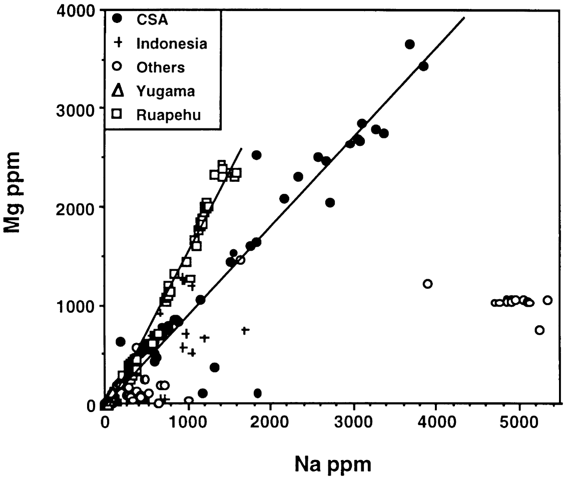

Fig. 3. Na versus Mg for the lake waters: both the Ruapehu and Poas (CSA) data sets show increases in both elements but their values are different.

saturation index calculations done on lake water analyses are not necessarily valid for the source fluids as well. Most common secondary silicate minerals are below saturation in the acid lake waters by several orders of magnitude. CO_2-rich lakes (e.g. lake Nyos) may be saturated with carbonates, such as siderite (Bernard and Symonds, 1989).

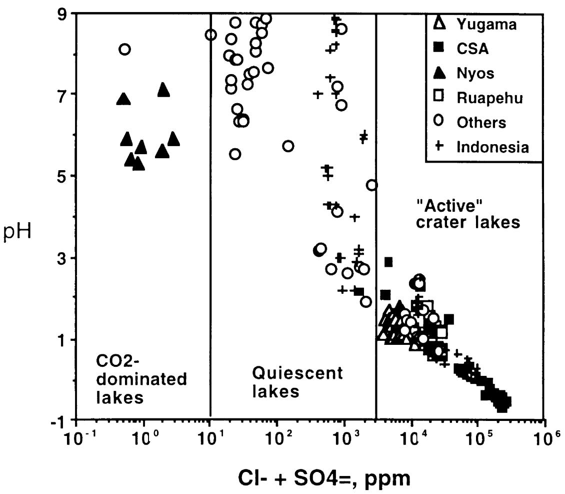

Fig. 4. Compositional ranges of volcanic lakes in terms of SO_4 + Cl and pH. Fields for CO_2-dominated lakes, quiescent geothermal lakes, and 'active' acid crater lakes (pH < 2) are shown.

4. Results of statistical analyses

Relations between chemical parameters in the data set are given by binary correlation coefficients (Table 2). Among anions, Cl, SO_4, and F are all strongly, positively correlated, whereas B shows a much weaker relationship with these. Among cations, 83% of the log–log correlation coefficients exceed 0.5 and half exceed 0.7. Between anions and cations, 92% of the log–log correlation coefficients exceed 0.5, while 58% exceed 0.7. The only elements that do not show strong correlations are Si, Li, and B, but all of these have a poor representation in the database.

Anion concentrations are positively correlated with RFE concentrations, especially in Poas and Yugama data. Most RFEs in volcanic lake water are derived from WR interaction, although contributions of volatile metal chlorides from high-temperature volcanic gases (Symonds et al., 1994) and concentrated volcanic brine inputs (Poorter et al., 1989; Lowenstern, 1994) may provide an additional input of RFEs in some systems. Inter-element correlations are probably the result of variable degrees of dilution between a fluid input with more or less fixed composition and seasonal or secular trends in the rate of dilution. Ruapehu fluids show more complex relations, with little variation in anion concentrations as cation concentrations increase (Fig. 1) and increases in Mg with little or no increase in Fe (Fig. 2).

The Na/Mg ratios vary within narrow margins for the Ruapehu data set (Fig. 3) suggesting that the compositional evolution of the Ruapehu source fluids over time is a result of varying degrees of WR interaction and WR ratios (Christenson and Wood, 1993).

Total dissolved solid concentrations in volcanic lakes correlate significantly and positively with all variables, except B (Table 2), and are very strongly predicted by SO_4 and Cl (Table 2). The TDS–anion relations indicate two distinct regions: for Cl + SO_4 greater than 200 ppm, lakes are acidic and TDS is dominated by volcanic acids. Neutral lakes are divided into CO_2-dominated ones such as Nyos having Cl + SO_4 less than 5 ppm and geothermal ones such as Batur with hydrothermal spring inputs having Cl + SO_4 between 5 and 200 ppm. High TDS lakes tend to be the hottest and most acidic, and these relations are portrayed in a pH versus Cl + SO_4 graph (Fig. 4).

Table 3
Unrotated principal components matrix with HCO_3, Mn, and Li excluded. The lakes represented in this analysis are Crater Lake on Mt Ruapehu (60% of data), several Indonesian lakes (Keli Mutu lakes, Egon Lake, Wai Sano, Kawah Putih, Kawah Ijen, Kelut, Lake Batur), El Chichon, Oyunuma, Bannoe, Popocatepetl, Maly Semiachik, and Rincon de la Vieja

	Principal components		
Variables	PC1	PC2	Communalities
T	0.38	0.73	0.51
H^+	0.84	−0.36	0.96
TDS	0.98	−0.08	0.99
Cl^-	0.95	0.17	0.98
SO_4	0.87	−0.37	0.97
F^-	0.85	−0.28	0.87
B	0.75	0.11	0.93
Na	0.71	0.6	0.96
Mg	0.86	0.25	0.96
Al	0.89	−0.36	0.97
SiO_2	0.67	0.32	0.79
K	0.84	0.23	0.95
Ca	0.86	−0.05	0.94
Fe	0.91	−0.29	0.95
Variation explained (%)	68	12	
Cumulative (%)	68	80	

When PCA was carried out using all constituents except HCO_3, Mn, and Li, there were 107 complete database entries. Two principal components were found in this run, and these explained 80% of data variability (Table 3). Communalities were found to exceed 95% for most variables, with the notable exceptions of temperature, SiO_2, and F. All variables except temperature were best represented by the same principal component (PC1). Because the loadings for Na were nearly equally distributed between the two components, rotating the axis led to it switching to PC2. When data for F, B, and SiO_2 also were excluded (missing in many analyses), the number of complete entries available for PCA increased to 157. Once again, two principal components were found, and in this case they explained 85% of data variability. Communalities were again high for most variables, with the exception of temperature and Ca. Unrotated loadings for all variables except temperature were again strongest in PC1. After rotation, Na, Mg, and K loadings were greater in PC2 than PC1.

The principal component analysis shows that most constituents have a strong coherence, suggesting a single underlying source. The anions Cl, SO_4, and F as well as the cations H, Al, and Fe all co-vary and are due to the same underlying principal component. Communalities for these ions show that virtually all of their variability can be explained by that one component that we suggest is representative of the volcanic gas input (H, S, Cl and F). While Al and Fe are not major constituents of the volcanic gas input, their abundances are linked through the strongly pH-dependent solubility properties of these two elements. The strong co-variation observed between F and Al is probably related to the occurrence of Al–F complexes that form in acidic solutions. The temperature and the cations Na, Mg, and K strongly co-vary with the main principal component, but some of their variability is explained by a second component. This component may reflect the temperature dependency of the Na/K ratio in less acidic fluids and indeed, a positive correlation exists between this second component and pH. When loadings are rotated, pH is no longer represented by it. These results indicate that the second principal component represents a geothermal input with variable temperature and a pH in the range 4–7. Finally, SiO_2 and Ca are represented by the main principal component, but they have the lowest communalities of all chemical variables. The two most common minerals that are saturated in many volcanic lakes are gypsum and silica, and we infer that the precipitation process

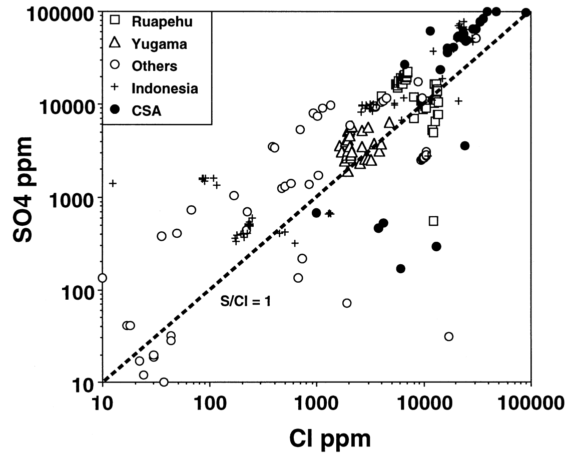

Fig. 5. SO_4 versus Cl graph, showing a mean value of S/Cl ~ 1 with large local variations.

distinguished Ca and Si from the others; the SO_4 concentration of most lakes is too high to be influenced by gypsum precipitation.

In summary, statistical analysis of the database indicates that lake fluid compositions may be influenced by three main processes:

(1) The strength of the volcanic gas input (H, Cl, S F) and related concentrations of some pH-dependent element solubilities (Al, Fe) which are broadly coupled with the degree of WR interaction expressed by amounts of main rock-forming cations.

(2) An input of a moderate pH, mineralized component (geothermal fluid) with a much weaker linkage between pH and cation concentrations than in 1.

(3) The extent of mineral precipitation, mainly calcium sulfate and silica. In addition, dilution and evaporation processes have their influence on the absolute concentrations, but do not impact element–element ratios.

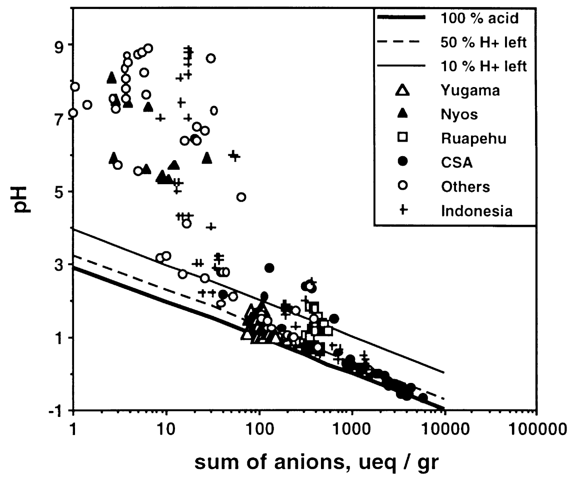

Fig. 6. Sum of anions (in micro-equivalents/g) versus measured pH values. Degree of neutralization isopleths for 50 and 90% are indicated. High pH waters have all high degrees of neutralization (small fraction of original H^+ left), whereas the most acidic waters have lost <10% of their original acidity through WR interaction.

5. Volcanic acid consumption

The source of acidity in volcanic lakes can be CO_2, be it volcanic or derived from decarbonation reactions in the underlying rocks. Dilute meteoric lake waters gain acidity by solution of atmospheric or deep CO_2 or by release of CO_2 during decay (oxidation) of organic matter. Other sources of acidity are the oxidation of pre-existing sulfides or sulfur, input of geothermal H_2S, or input of volcanic gases. Subsequent hydrolysis reactions consume H^+, increasing solution pH, and the concentration of bicarbonate and cations in lake waters. In lakes with a volcanic gas input rich in SO_2 and the strong acids HCl and HF, acidity derived from CO_2 becomes negligible. Distinct thresholds exist for CO_2-dominated lakes, dilute geothermal lakes, and hot acid–brine lakes in a graph with lake water pH and the sum of Cl + SO_4 (Fig. 4).

A graph with Cl versus SO_4 shows lakes dominated by either H_2S input or oxidation of sulfur/sulfides and lakes with an active volcanic brine input (Cl and S rich). The variation in volcanic lake S/Cl depends on the original volcanic gas (H_2S + SO_2)/HCl and on the fate of volcanic SO_2 in the hydrothermal/lake system. Volatilization of previously deposited native sulfur when fluid temperatures exceed the boiling point of sulfur may add sulfur as well (Christenson, 2000 – this volume). The disproportionation of SO_2 into either $H_2S + HSO_4^-$ or $S^0 + HSO_4^-$ at pH values <2 (Kusakabe et al., 2000 – this volume) will have a strong impact on the dissolved S/Cl. H_2S may escape as a gas at the lake surface, whereas S^0 may form a commonly molten sulfur layer in the volcano-hydrothermal system and/or on the lake bottom (e.g., Takano et al., 1994). In addition, Cl may escape as HCl vapor at the lake surface in hot lakes like Poas

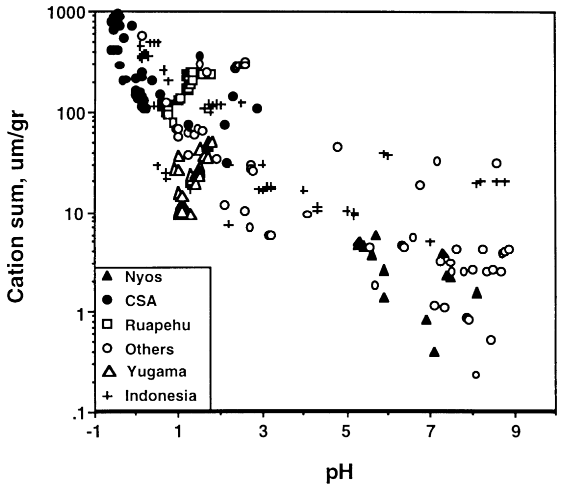

Fig. 7. Relation between pH and total cation concentrations: the most acidic fluids are also most concentrated in major rock forming elements.

(Rowe et al., 1992a). Average arc volcanic S/Cl values, measured on high-temperature fumaroles, show a wide range (0.1–10; Symonds et al., 1994), indicating that observed lake S/Cl values cannot be simply compared to a 'standard' arc volcanic S/Cl value. The most active Crater Lakes (e.g. Poas, Ruapehu, Yugama) have S/Cl ~ 1 (Fig. 5), be it with substantial variation. Evidence for the presence of liquid sulfur in these systems suggests that the volcanic input fluids had larger S/Cl. The stoichiometry of the S^0 precipitation reaction (Kusakabe et al., 2000 – this volume) suggests that S/Cl ~ 2 is an approximation of their average value.

We use here 'degree of neutralization' as a new parameter to trace the WR interaction history of volcanic lake fluids. The 'sum of anions' versus pH graph (Fig. 6) shows the theoretical pure acid boundary (no WR interaction) and calculated H^+ depletion isopleths that represent, respectively, 50% neutralization and 90% neutralization (10% of original acidity left). The distribution of data points shows a slanted array, with the most dilute lakes more than 90% neutralized, fluids with a pH of 1–2 between 50 and 90% neutralized (with some exceptions) and the most acidic fluids are all <50% neutralized. Poas fluids and some Yugama and Indonesian samples plot close to the pure acid end-member, suggesting the presence of a quenched volcanic brine that has only seen minor amounts of WR interaction. Despite their low pH values, many Ruapehu fluids are >50% neutralized. The absolute amounts of H^+ consumed in the very acid fluids are much larger than those in the neutral

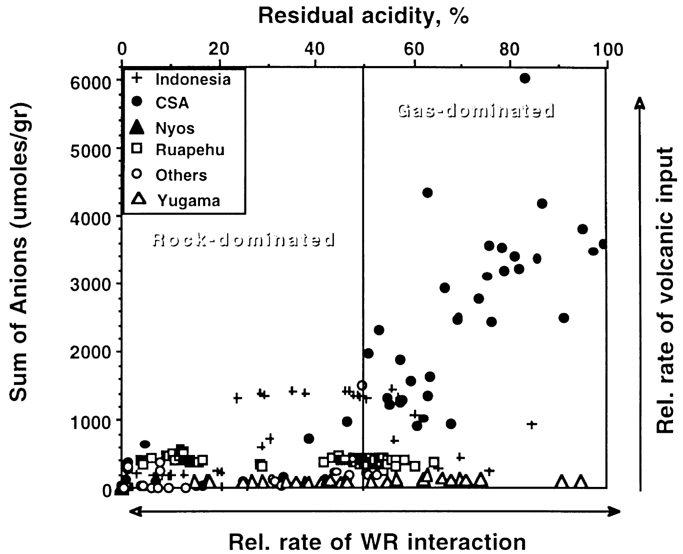

Fig. 8. Fields of gas dominated versus rock-dominated crater lake fluids, based on degree of neutralization (expressed as % residual acidity) and sum of the anion concentration (measure of volcanic gas input). The vertical axis is a measure of rate of volcanic gas input (and degree of dilution or evaporation) and the horizontal axis displays the reactivity of the host rocks of the hydrothermal system.

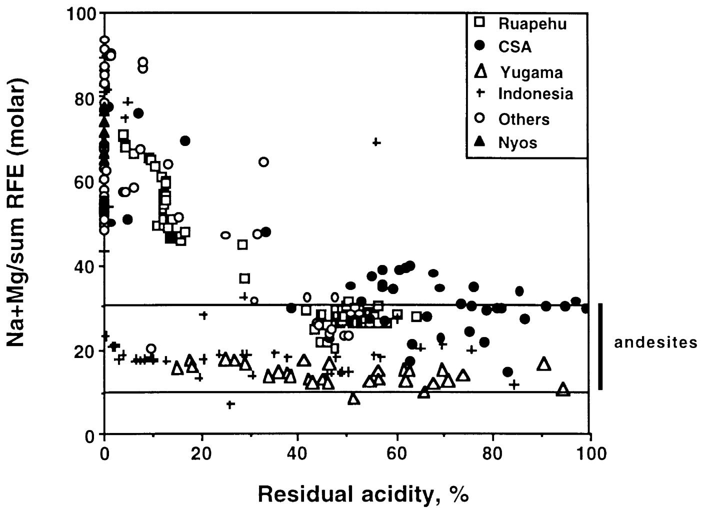

Fig. 9. The relative abundance of Na + Mg in the cation balance versus the PRA parameter. Fluids that are most extensively neutralized have the highest relative Na + Mg abundance. In these strongly neutralized fluids, Na and Mg are linearly and positively correlated. Thin lines indicate the compositional field of common andesites.

pH fluids, even for those very acid fluids with low degrees of neutralization. A graph of pH versus ΣRFE shows the highest absolute cation concentrations for the lowest pH fluids (Fig. 7), which in addition to WR interaction is also influenced by evaporation. The 'amount of H^+ consumed' is the vertical distance between a data point and the pure acid boundary in Fig. 6 and the percentage residual acidity or PRA used here is defined by:

$$\frac{100\%\{(\text{pure acid H}^{+}\text{level}) - (\text{amount of H}^{+}\text{consumed})\}}{(\text{pure acid H}^{+}\text{level})}.$$

This parameter, which is independent of dilution or evaporation, is the complement to the degree of neutralization and is almost 100% for data near the pure acid end-member, and close to zero for almost fully neutralized samples. *Process rates* are of importance for these very dynamic hydrothermal systems and lakes, and the degree of neutralization is a measure of the relative rate of volcanic gas input (acidification) versus the relative rate of WR interaction (neutralization) (Fig. 7) The former is also indicated by the absolute anion concentrations in the fluids, indicating the degree of dilution or an increase in the gas input rate. Of course, evaporation would also increase the anion contents, but H^+ versus anion plots for Poas lake do not show linear evaporation trends. We define two regions in a graph of PRA versus the sum of the anions: gas-dominated lake waters, with >50% of original acidity retained and little dilution, and rock-dominated waters, with <50% of the original acidity retained and commonly higher levels of dilution (Fig. 8). Lake waters will plot high on the right side of the diagram when the magnitude of the volcanic gas input is large (and evaporation has an impact as well) and low on the right side when the rate of neutralization is small and dilution occurs. Hydrothermal systems with already strongly altered rocks and a lack of easily leached cations may display such low rates of neutralization. Data on the left side of the diagram come from lakes with either small volcanic gas inputs and/or a very reactive rock matrix (e.g. hot, freshly erupted magma).

Data collected at Laguna Caliente at Poas volcano from 1985 to 1991 represent a period when the lake was affected by a high gas flux with very little dilution such that the rate of acidification strongly exceeded the rate of WR interaction. Volcanic material beneath Laguna Caliente is likely to have been extensively altered as the nearly continuous existence of a highly mineralized volcanic lake in the active crater of Poas was documented well over a century ago by European

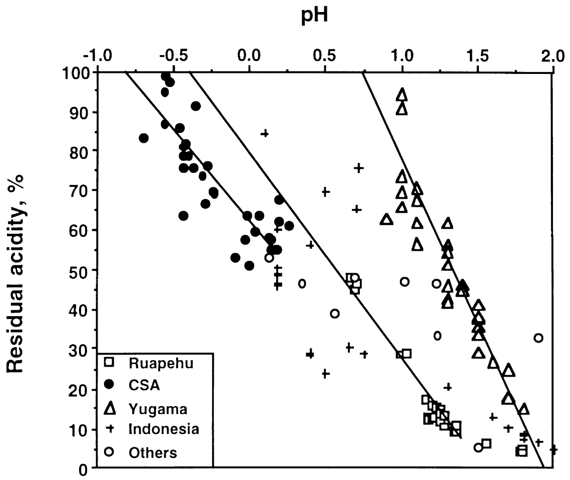

Fig. 10. Measured pH values are plotted versus PRA retained. The pH of the original input fluids can be crudely constrained by extrapolation of the individual data arrays (Ruapehu, Poas and Yugama) to the 100% acidity retained boundary.

explorers (Von Frantzius, 1861). Many samples from Crater Lake of Mt. Ruapehu have compositions typical of a rock-dominated system, with water chemistry reflecting interaction between chilled fresh magma and acid fluids, whereas water samples taken during the more active 1995–1996 period plot in the gas-dominated field. Yugama displays the full array of partial neutralization values over time, but the fluids are never very concentrated. Possibly, lake water is recycled through the underlying geothermal system with small inputs of a new volcanic fluid (Ohba et al., 1994) and the rock matrix may already be strongly altered as well. No recent magmatic eruptions are known from Yugama. The data from Indonesian volcanic lakes plot partly in the gas-dominated field (e.g. Keli Mutu TiN and Kawah Ijen) or in the rock-dominated systems (e.g. Kelut and Wai Sano). The highly acidic volcanic lake of Copahue volcano, Argentina plots in the gas-dominated field. The horizontal axis of Fig. 8 roughly correlates with the availability of fresh volcanic rocks to neutralize volcanic acids, whereas the vertical axis relates to the magnitude of volcanic gas input and/or degree of dilution/evaporation. PRA versus the relative amount of Na + Mg shows that the most neutralized fluids have the highest relative Na + Mg concentrations (Fig. 9). All samples that are <80% neutralized have similar relative Na + Mg concentrations. This indicates that interactions of very acid fluids with fresh volcanic rocks initially leach much of the Na and Mg, and the fluids become neutralized in the process. Such neutralized fluids have seen only very superficial interaction with fresh volcanic rocks. Further attack of the silicate framework of the rock matrix by subsequent batches of acid fluids leads to release of Al and Si, which is a substantially slower process and leads to lower degrees of neutralization and lower relative Na + Mg concentrations. Alternatively, the fluids with high PRA values (and very low pH) have also high levels of Al and Fe, creating low Na + Mg/ΣRFE, whereas the low PRA fluids have lost most of their Al and Fe in secondary minerals like alunite and jarosite. Such fluids also may have equilibrated with common Al-silicate secondary minerals early on in their history but subsequently mixed with new volcanic acids. Such fluid compositions, with a depletion in some RFE relative to the rock protolith, are no longer saturated with secondary Al-silicate minerals.

We gain some insight into the nature of the source fluids by plotting the lake-water pH versus PRA (Fig. 10). The data plot in separate arrays for the three main data series (Poas, Ruapehu, Yugama) and a scatter of data from the many other lakes. We extrapolated somewhat arbitrarily in a linear fashion the three main data arrays back to their most primitive fluids (PRA = 100) to obtain their possible source compositions. This procedure removes the effects of WR interaction, but incorporates the local dilution effects. Source fluids with a pH = −0.8 for Poas, −0.4 for Ruapehu and +0.75 for Yugama were found. Volcanic gases without much dilution may be entering Poas and Ruapehu lakes, whereas much more dilute fluids enter the Yugama system. Modeling of the condensation of a volcanic gas gives pH values of only 0.8–1 (Rowe, 1991; Delmelle and Bernard, 1994), but further oxidation of that fluid or disproportionation of SO_2 may lead to such low-pH condensed gas fluids. The degree of dilution of the source fluids can be an effect of the local hydrology or may be controlled by the magnitude of the volcanic gas flux. Lake-water temperatures suggest that the latter is the most likely explanation for these three lakes.

The various types of WR interaction can be traced on a diagram of absolute amount of H^+ used versus

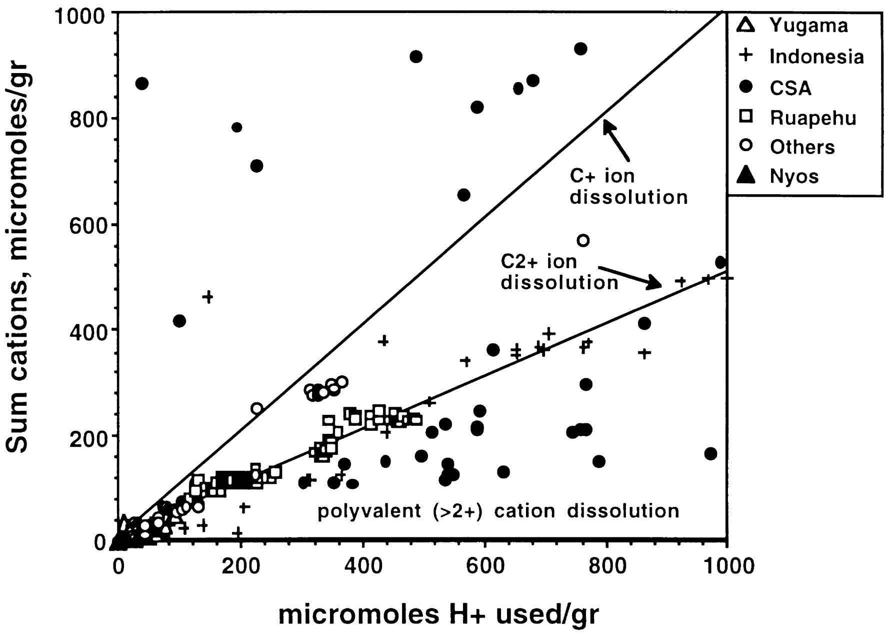

Fig. 11. During water-rock interaction, H^+ is exchanged for cations, maintaining charge balance. The sloping lines are the exchange reactions between H^+ for monovalent and for bi-valent cations. The field below the bivalent line contains fluids rich in trivalent cations. The field above the monovalent line has acquired cations apparently without H^+ exchange.

ΣRFE (Fig. 11). We can interpret trends on this diagram in terms of incomplete rock dissolution (leaching) or in terms of congruent rock dissolution with or without subsequent precipitation of secondary phases. We indicated lines of conversion of H^+ to cations with different valences: e.g. when only Na^+ is released, one H^+ is used for one Na^+ and so on. The mean valence of RFEs in solid rocks is close to 2^+ because the main rock anion is O^{2-}. Fluids that have experienced congruent rock dissolution thus should straddle the 2^+ conversion line. No fluid compositions should occur in the field above the H^+ to univalent cation boundary, although evaporation and dilution will create trends along slope = 1 lines from their starting composition. Fluids that have congruently dissolved rocks and then lost part of their cation content because of precipitation of hydrous sulfates (e.g. alunite, jarosite) would plot below the 2^+ line and precipitation of oxides like Fe_3O_4 or kaolinite would give compositions that plot above the 2^+ line. The effects of precipitation of minerals with 2^- anions (e.g. anhydrite) after congruent dissolution cannot be detected in this diagram. It is unclear how the samples above the 1^+ conversion line originated (mainly Poas samples); possibly the errors introduced with neglecting activity coefficients for H^+ have an impact here. Most of the Ruapehu samples straddle the $2H^+$ to bivalent cations boundary, suggesting congruent dissolution of rocks and/or selective dissolution of minerals such as albitic feldspar ($4H^+$ for $Na^+ + Al^{3+}$). The fluids that occur at the bottom of the diagram may have dissolved remnant Si–Al-rich phases which lead to Al-enrichment. Forward WR interaction modeling with a program like Chiller (Reed, 1982) should be done to model the occurrence of fluid compositions within the different fields of this diagram and test its usefulness.

6. Secondary minerals

Many of the acid, concentrated lake waters in the database are saturated with one or more silica phases (amorphous silica, cristobalite or quartz) and gypsum and barite. No secondary silicate minerals like clays or zeolites are saturated in any of the acid lakes in the

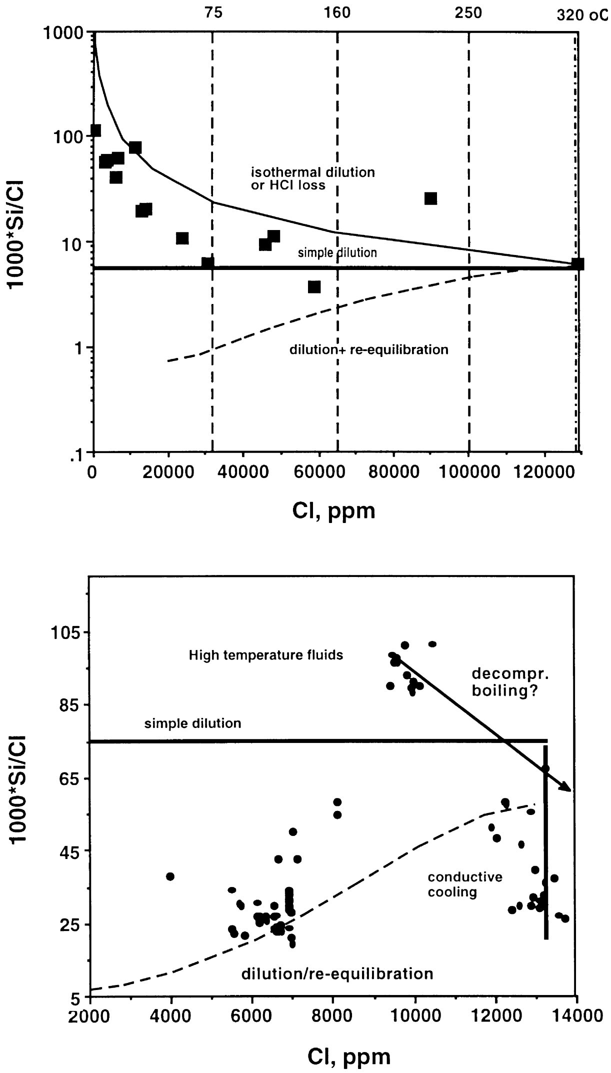

Fig. 12. The Cl–Si graphs show the various modes of cooling for hydrothermal fluids. Conservative mixing between cold (meteoric) and hot mineralized fluids produces linear, horizontal cooling–dilution relations (simple dilution). The lower curved line represents mixing (= cooling) with re-equilibration of Si in the diluted fluids (after Fournier, 1985). The upper curved line represents mixing between deep hot mineralized fluids and hot surficial fluids with identical Si concentrations but no Cl or HCl loss from the mixed fluids at the lake surface. Vertical lines indicate conductive cooling with silica re-equilibration. (a) Results for Poas, with evidence for rapid fluid transfer and little silica re-equilibration; and (b) results for Ruapehu with evidence for slow cooling and partial silica re-equilibration.

database. The most concentrated lakes are also saturated with native sulfur, which may accumulate at the lake floor or may form a floating sulfur froth. Most lakes became silica saturated as a result of an input of hot fluids with high silica concentrations that mixed with the cool lake waters, and not as a result of in situ WR interaction. Lake-water temperature is strongly influenced by surface heat exchange (e.g. Pasternack and Varekamp, 1997), and thus bears no simple relationship with the temperature of the original input fluid. Temperature and silica concentration of the input fluid are related by a temperature-dependent equilibrium (Fournier, 1985), and correlations between a largely conservative fluid input tracer (e.g. Cl) and SiO_2

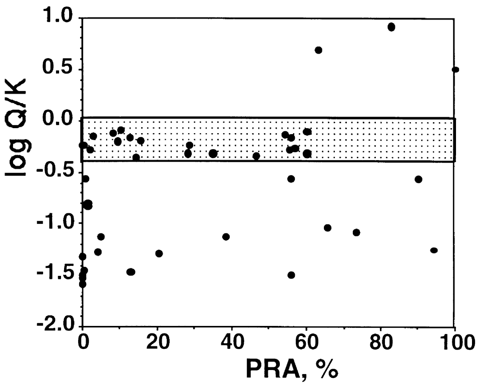

Fig. 13. Gypsum saturation as a function of % acidity retained (PRA). The log *Q*/*K* values indicate the degree of gypsum saturation at the lake temperature as calculated by SOLVEQ. Only fluids with >40% neutralization (>60% PRA) are near saturation with gypsum, indicated by the dashed band.

concentrations may be used to examine mixing and cooling relations. The rate of fluid throughput can be estimated from these diagrams by assuming that precipitation kinetics of silica in the relatively low-temperature lake environment are slow enough to approximate conservative aqueous silica behavior.

We investigated Si–Cl relations for Ruapehu and Poas in detail, because these systems had the highest silica concentrations. The most saline and silica-rich fluid samples, which probably represent the least diluted fluids from the underlying reservoir, give a quartz equilibration temperature of about 320°C for both volcanic lake systems. These hot fluids can slowly cool by conduction (Cl^- remains constant, silica precipitates), mix with cold fluids without re-equilibration (Si/Cl^- remains fixed), mix with cold fluids and re-equilibrate at the new temperatures (Cl^- decreases, silica precipitates) or isothermal mixing with hot meteoric fluids (Cl^- decreases, Si remains constant), as shown in Fig. 12.

In the Ruapehu data we can distinguish one concentrated hot end-member as well as a group of samples that were hotter but less Cl-rich. The latter may constitute an ultimate source fluid that generated more saline fluids through decompressional boiling with associated cooling. From these very saline fluids evolves a cooler group through conductive cooling and a group of samples that plots between simple dilution and dilution with re-equilibration. Floating silica precipitates are indeed common in Ruapehu Crater Lake (B.W. Christenson, pers. commun. 1999). Most of these fluids have traveled relatively slowly and partially re-equilibrated during and after mixing. Poas data plot largely in the field between simple dilution and some form of isothermal mixing, suggesting a fluid that traveled rapidly through a system and possibly mixed with a hot, Cl poor thermal fluid with similar Si concentrations. Such mixing could have been caused by the geyser-like phreatic eruptions that occurred when many of the Poas samples were collected. These eruptions are caused by boiling at depths 50–150 m below the lake bottom and would cause mixing of hot, silica-rich fluids with cooler lake brines (Casertano et al., 1987). Alternatively, at the time Poas samples were collected, conditions of extreme acidity and elevated lake temperature were causing HCl vapors to be released from the lake surface and non-conservative behavior of Cl in lake brine was observed (Rowe et al., 1992b). This would also explain the trend of decreasing Cl with constant silica.

The Si–Cl relations may thus serve as a hydrothermal speedometer, indicating the relative rates of re-equilibration during cooling and dilution. Yugama and many Indonesian lakes have much cooler and dilute input fluids, as indicated by the lower Si/Cl ratios and lower absolute Si concentrations.

Many lakes are close to saturation with gypsum, and the situation is here more complex than for silica. Gypsum saturation in these sulfate-rich acidic fluids is temperature and pH dependent because of the $HSO_4{}^-$/$SO_4{}^=$ equilibrium. We calculated gypsum saturation indices with SOLVEQ (Reed, 1982) and plotted log *Q*/*K* values against the PRA parameter (Fig. 13). Only lakes with PRA < 60% are near saturation with gypsum, but many fluids with high levels of neutralization are gypsum undersaturated. This could be an indication of dilution, because the PRA parameter is dilution insensitive, whereas *Q* is dilution sensitive. The database has no fluids that have only reacted with a barren rock protolith with anhydrite already present—this would potentially provide fluids with very low degrees of neutralization but at gypsum saturation.

7. A geochemical classification approach

Grouping of volcanic lakes can be accomplished through cluster analysis and with the help of some reasoning discussed above. When HCO_3, Mn, Li, B, F, and SiO_2 were excluded from the database, 157 entries had measures for the remaining 11 ions. Based on these 157 entries, cluster analysis shows several distinct levels of classification (Fig. 14). At the highest level, lake-water geochemistry delineates neutral lakes from acid lakes, as indicated by the high linkage distance between these groups. Acid volcanic lakes show greater geochemical similarities as a whole than neutral volcanic lakes. Among neutral lakes, CO_2-rich volcanic lake samples are strongly differentiated from the others; that is partly caused by their very low Cl and SO_4 concentrations but also reflect their low concentrations of other constituents. The remaining neutral lakes are all quite distinct because of variations in their hydrothermal inputs and different country rocks that host these systems and different fluid temperatures.

Two groups can be distinguished in the acid lake group: acid–sulfate waters with TDS less than 1% and some lakes with TDS between 1 and 4%. The group I acid samples shows greater overall similarity than the neutral lakes, but less than the lakes with higher TDS. The differences among these lakes relate to small pH and temperature variations, which translate in substantial differences in Ca and Fe. The second group of acid lakes derived from the cluster analysis contains samples from acid–saline lakes (TDS between 1–4%) and acid–brine lakes (TDS greater than 4%). Most of the samples from Ruapehu as well as those from Yugama, Maly Semiachik, Poas, Keli Mutu TAP, and Sirung have TDS of 1–4%. Ruapehu and Yugama were classified as acid–brine lakes (Pasternack and Varekamp, 1997) because these lakes are frequently injected with hot acid–brines and have subaqeuous molten sulfur pools. When considering the complete aqueous geochemical composition in

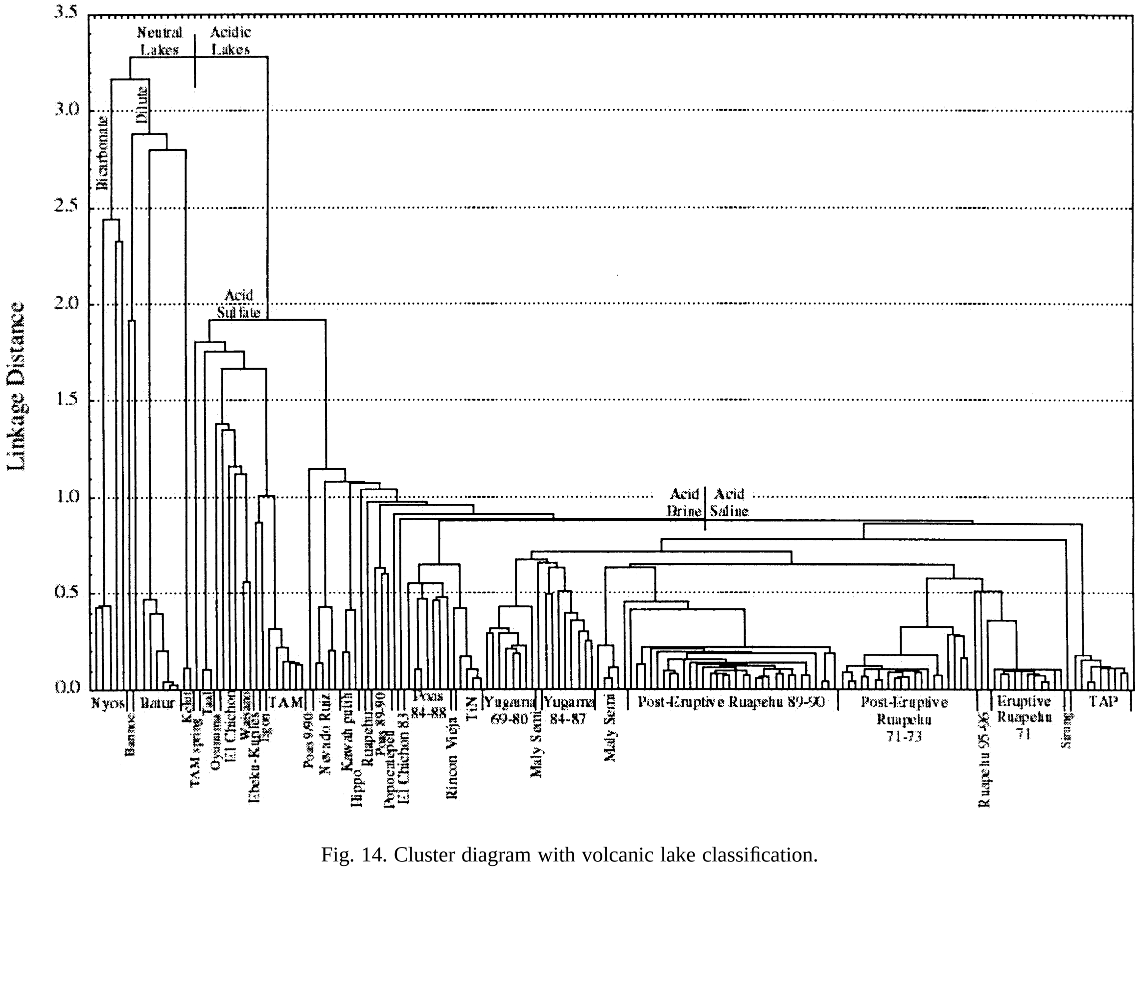

Fig. 14. Cluster diagram with volcanic lake classification.

a cluster analysis, these two lakes show the strongest grouping of all, and fit best within the acid–saline realm. Within Ruapehu and Yugama samples, subtle differences in overall geochemistry result in different clusters of lake samples. Among acid–brine lakes, Keli Mutu TiN, Poas, Kawah Ijen and Rincon de la Vieja are tightly grouped.

8. Conclusions

This paper is an initial, rather primitive attempt at a global assessment of crater lake chemical compositions, which does not do justice to the many excellent and detailed temporal studies of individual lakes. Statistical analysis shows widespread similarities and differences among volcanic lakes. Lakes are first distinguished by their main acidifying agents—CO_2, various S-bearing compounds, or HCl + HF + H_2SO_4. When CO_2 is the acidifying agent, bivariate correlations and cluster analysis show that lakes are either bicarbonate-dominated or subjected to local geothermal inputs that contain cations derived from WR interaction with surrounding rocks. When HCl + H_2SO_4 are the main acidifying agents, a single sublimnic source dominates, with cations from WR interaction increasing as a function of increasing amounts of anions from volcanic gases. Principal component analysis indicates the presence of a second source of Na, Mg, and K, likely from warm to hot, intermediate pH geothermal inputs.

The mainly sublimnic processes that cause differences between the lake groups can be summarized as follows: the flux of high-temperature volcanic gases versus the degree of neutralization, and the rate of fluid throughput in the underlying system. The latter may be qualitatively traced by the degree of dilution, relative Na and Mg abundances, and Si–Cl relations. End-member acid lake types defined by these two processes are:

(1) Volcanic gas-dominated systems, with a strong volcanic gas input, low degree of neutralization and probably a highly altered rock matrix, with a short residence time of the fluids in the hydrothermal system. These are the acid–brine type systems, where the rate of acidification greatly exceeds the rate of neutralization and fluids circulate relatively rapidly throughh the sublimnic hydrothermal systems.

(2) Rock-dominated systems, where WR interaction dominates the fluid composition, the residence time of the fluids in the hydrothermal system is longer, and the degrees of neutralization and dilution are significantly higher. These are the acid–saline type of lakes, where the rate of neutralization keeps up or exceeds the rate of acidification.

Volcanic lakes with a direct volcanic gas input are rare, except prior or during volcanic eruptions (e.g., Ruapehu, 1995–1996). Even many of the most strongly acidified lakes lack field evidence for input of a high-temperature gas, as suggested by the absence of a CO_2-rich gas plume, possibly with the exception of Keli Mutu lake TAP. Many lakes have an input of a volcano-hydrothermal component which resulted from the interaction of a volcanic gas with a hydrothermal system below the lake.

The lake systems can be seen as the tops of the volcano-hydrothermal systems, with their chemical composition largely determined by the flux and composition of the deep fluid input. The scheme in Fig. 14 provides a current overview of the classification of volcanic lake fluid compositions.

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References

Badrudin, M., 1994. Kelut volcano monitoring: hazards, mitigation and changes in water chemistry prior to the eruption. Geochem. J. 28, 233–242.

Bernard, A., Symonds, R.B., 1989. The significance of siderite in the sediments of Lake Nyos, Cameroon. J. Volcanol. Geotherm. Res. 39, 187–194.

Brantley, S.L., Borgia, A., Rowe, G., Fernandez, J.F., Reynolds, J.R., 1987. Poás volcano volcanic lake acts as a condenser for acid metal-rich brine. Nature 330, 470–472.

Delmelle, P., Bernard, A., 1994. Geochemistry mineralogy and chemical modeling of the acid volcanic lake of Kawah Idjen volcano, Indonesia. Geochim. Cosmochim. Acta 58, 2445–2460.

Casertano, L., Borgia, A., Cigolini, C., Morales, L.D., Gomez, M., Fernandez, J.F., 1987. An integrated dynamic model for volcanic activity at Poas Volcano, Costa Rica. Bull. Volcanol. 49, 588–598.

Chiodini, G., Cioni, R., Leonis, C., Marini, L., Raco, B., 1993. Fluid geochemistry of Nisyros island, Dodecanese, Greece. J. Volcanol. Geotherm. Res. 56, 95–112.

Christenson, B.W., 1994. Convection and stratification in Ruapehu Crater Lake, New Zealand: implications for Lake Nyos-type gas release eruptions. Geochem. J. 28, 185–199.

Christenson, B.W., 2000. Geochemistry of fluids associated with the 1995/96 eruption of Mt. Ruapehu, New Zealand: signatures and processes in the magmatic–hydrothermal system. J. Volcanol. Geotherm. Res. 97, 1–30.

Christenson, B.W., Wood, C.P., 1993. Evolution of a vent-hosted hydrothermal system beneath Ruapehu Crater Lake, New Zealand. Bull. Volcanol. 55, 547–565.

Evans, W.C., Kling, G.W., Tuttle, M.L., Tanyileke, G., 1993. Gas buildup in lake Nyos, Cameroon: the recharge process and its consequences. Appl. Geochem. 8, 207–221.

Fournier, R., 1985. The behavior of silica in hydrothermal solutions. In: Berger, B.R., Henley, R.W. (Eds.), Geology and Geochemistry of Epithermal Systems. Soc. Econ. Geol. Rev., Econ. Geol. 21, 45–61.

Giggenbach, W.F., 1974. The chemistry of crater lake, Mt. Ruapehu (New Zealand) during and after the 1971 active period. N.Z. J. Sci. 17, 33–45.

Giggenbach, W.F., 1983. Chemical surveillance of active volcanos in New Zealand. In: Tazieff, H., Sabroux, J.C. (Eds.). Forecasting Volcanic Events. Elsevier, Amsterdam, pp. 311–322.

Giggenbach, W.F., 1988. Geothermal solute equilibria Derivation of Na–K–Mg geoindicators. Geochim. Cosmochim. Acta 52, 2749–2765.

Giggenbach, W.F., 1990. Water and gas chemistry of Lake Nyos and its bearing on the eruptive process. J. Volcanol. Geotherm. Res. 42, 337–362.

Giggenbach, W.F., Glover, R.B., 1975. The use of chemical indicators in the surveillance of volcanic activity affecting the volcanic lake on Mt. Ruapehu, New Zealand. Bull. Volcanol. 39, 132–145.

Hair Jr., J.F., Anderson, R.E., Tatham, R.L., Black, W.C., 1992. Multivariate Data Analysis, 3rd edition, Macmillan, New York.

Kusakabe, M., Ohsumi, T., Aramaki, S., 1989. The Lake Nyos gas disaster: chemical and isotopic evidence in waters and dissolved gases from three Cameroonian crater lakes Nyos, Monoun, and Wum. J. Volcanol. Geotherm. Res. 39, 167–185.

Kusakabe, M., Komoda, Y., Takano, B., Abiko, T., 2000. Sulfur isotopic effects in the disproportionation reaction of sufur dioxide in hydrothermal fluids: Implications for the δ^{34}S variations of dissolved bisulphate and elemental sulfur from active crater lakes. J. Volcanol. Geotherm. Res. 97, 287–307.

Lowenstern, J., 1994. Chlorine, fluid immiscibility and degassing in peralkaline magmas from Pantellaria. Am. Mineral. 79, 353–369.

Martinez, M., Fernandez, E., Valdes, J., Barboza, V., Van der Laat, R., Sandoval, L., Duarte, E., Malavassi, E., Barquero, J., Valverde, J., Marino, T., 2000. Chemical evolution and volcanic activity and of the active crater lake of Poas Volcano, Costa Rica, 1993–1997. J. Volcanol. Geotherm. Res. 97, 127–141.

Nordstrom, D.K., Alpers, C.N., 1998. Negative pH, efflorescent mineralogy, and consequences for environmental restoration at the Iron Mountain Superfund site, California. Proc. Nat. Acad. Sci. 96, 3455–3462.

Ohba, T., Hirabayashi, J., Nogami, K., 1994. Water, heat, and chloride budgets of the volcanic lake Kusatsu-Shirane volcano, Japan. Geochem. J. 28, 199–216.

Ohba, T., Hirabayashi, J., Nogami, K., 2000. D/H and $^{18}O/^{16}O$ ratios of water in the crater lake at Kusatsu-Shirane volcano, Japan. J. Volcanol. Geotherm. Res. 97, 329–346.

Pasternack, G.B., Varekamp, J.C., 1994. The geochemistry of the Keli Mutu crater lakes, Flores, Indonesia. Geochem. J. 28, 243–262.

Pasternack, G.B., Varekamp, J.C., 1997. Volcanic lake systematics. I.: physical constraints. Bull. Volcanol. 58, 528–538.

Poorter, R.P.E., Varekamp, J.C., Van Bergen, M.J., Kreulen, R., Sriwana, R., Vroon, P.Z., Wirakusumah, A.D., 1989. The Sirung boiling spring: an extreme chloride-rich acid brine on Pantar (Lesser Sunda Islands, Indonesia). Chem. Geol. 76, 215–228.

Reed, M.H., 1982. Calculation of multicomponent chemical equilibria and reaction processes in systems involving minerals, gases, and an aqueous phase. Geochim. Cosmochim. Acta 46, 513–528.

Rowe, G.L., 1991. The acid crater lake system of Poas Volcano, central Costa Rica: Geochemistry, hydrology, and physical characteristics. The Pennslyania State University, University Park, PA, 309 pp.

Rowe, G.L., Brantley, S.L., Fernandez, M., Fernandez, J.F., Borgia, A., Barquero, J., 1992. Fluid-volcano interaction in an active stratovolcano: the volcanic lake system of Poás Volcano, Costa Rica. J. Volcanol. Geotherm. Res. 49, 23–51.

Rowe, G.L., Ohsawa, S., Takano, B., Brantley, S.L., Fernandez, J.F., Barquero, J., 1992. Using Crater Lake chemistry to predict volcanic activity at Poás Volcano, Costa Rica. Bull. Volcanol. 54, 494–503.

Rowe, G.L., Brantley, S.L., 1993. Estimation of the dissolution rates of andesitic glass, plagioclase and pyroxene in a flank aquifer of Poas volcano, Costa Rica. Chemical Geology 105, 71–87.

Rowe, G.L., Brantley, S.L., Fernandez, J.F., Borgia, A., 1995. The chemical and hydrologic structure of Poas Volcano, Costa Rica. J. Volcanol. Geotherm. Res. 64, 233–267.

Sanford, W.E., Konikow, L.F., Rowe, G., Brantley, S.L., 1995. Ground-water transport of crater-lake brine at Poas Volcano, Costa Rica. J. Volcanol. Geotherm. Res. 64, 271–297.

Sigurdsson, H., 1977. Chemistry of the crater lake during the 1971–72 Soufriere eruption. J. Volcanol. Geotherm. Res. 2, 165–186.

Sriwana, T.M., van Bergen, M.J., Varekamp, J.C., Sumarti, S., Takano, B., van Os, B.J.H., Leng, M.J., 2000. Geochemistry of the acid Kawah Putih lake, Patuha Volcano, West Java, Indonesia. J. Volcanol. Geotherm. Res. 97, 77–103.

Symonds, R.B., Rose, W.I., Bluth, G.J.S., Gerlach, T.M., 1994.

Volcanic-gas studies: methods, results, and applications. Rev. Mineral. 30, 1–66.
Takano, B., Watanuki, K., 1990. Monitoring of volcanic eruptions at Yugama crater lake by aqueous sulfur oxyanions. J. Volcanol. Geotherm. Res. 40, 71–87.
Takano, B., Saitoh, H., Takano, E., 1994. Geochemical implications of subaqueous molten sulfur at Yugama, Kusatsu-Shirane Volcano, Japan. Geochem. J. 28, 199–216.
Takano, B., Zheng, S., Oshawa, S., 2000a. A telemetering system for monitoring aqueous polythionates in an active crater lake. J. Volcanol. Geotherm. Res. 97, 397–406.
Takano, B., Fazlullin, S.M., Delmelle, P., 2000b. Analytical laboratory comparison of major and minor constituents in an active crater lake. J. Volcanol. Geotherm. Res. 97, 497–508.
Von Frantzius, A., 1861. A contribution to the knowledge of the volcanos of Costa Rica. In: Vargas, C.A. (Ed.), Antologia el volcan Poas. Universidad de Estatal a Distancia, San Jose, Costa Rica, pp. 11–34 in Spanish.